

Techno-economic optimization of power-to-methanol with co-electrolysis of CO₂ and H₂O in solid-oxide electrolyzers

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ABSTRACT

Power-to-methanol processes with co-electrolysis in solid-oxide electrolyzer provides a promising approach to deal with two problems: large-scale renewable electricity storage and carbon capture and utilization. In this paper, a large-scale power-to-methanol system with solid-oxide electrolyzer co-electrolysis technology is studied. System-level heat integration and techno-economic assessment are performed by using a multi-objective optimization platform. The results indicate that there is a slight trade-off between the overall energy efficiency and methanol production cost. The system can achieve a high energy efficiency (72%) and carbon conversion efficiency (93.6%), with the annual CO₂ utilization reaching 146.7 kton. However, its economic cost is high. SOE stack price, stack lifetime, and electricity price are crucial to the economic competitiveness of the system. By reducing the cost of SOE stack, extending its lifetime and reducing electricity price, the payback time can be shortened to 3–5 years. A stable supply of renewable electricity is necessary for the project investment. When the annual available hours of renewable electricity drop from 7200 to 3600, the payback time of the project increases to 21 years. Methanol synthesis with SOE co-electrolysis has an outstanding heat integration performance. The system can recover heat in a Rankine cycle to enhance the overall energy efficiency.

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1. Introduction

The ever-increasing consumption of fossil fuels, namely, oil, natural gas, and coal, has raised the level of CO₂ in the atmosphere, leading to undesirable climate change and environmental damage [1–3]. In order to reduce the use of fossil fuels, renewable energy sources (RESs) play a significant role, especially in the field of power generation. For example, China has set a target to increase the use of RESs to 27% by 2020 [4]; and Germany plans to generate 80% of its electricity from RESs by 2050 [5]. However, with increasing penetration of RESs (such as solar power and wind power), the dynamic and intermittent nature of most RESs and frequent mismatch of renewable-power supply and demand have to be dealt with. Consequently, large-scale renewable electrical energy storage (REES) and redistribution is necessary to continue with the installation of RESs connected to the electric grid. Beyond the electrical energy, many industrial processes (such as petrochemical, iron and steel, aluminum, paper and pulp, refineries, and cement) and the

fuels for terrestrial, marine and aeronautic propulsion in our modern society strongly rely on fossil resources, leading to huge amounts of CO₂ emissions [6,7]. As a result, an effective way is to capture and convert CO₂ into chemicals and fuels of commercial values instead of releasing it to the atmosphere [8].

High temperature solid-oxide electrolysis (SOE) is a promising technology to store renewable energy and convert CO₂ through the co-electrolysis of H₂O and CO₂ directly into syngas with a desired H₂ to CO ratio for the downstream process [9]. Compared with the low-temperature electrolysis technologies, SOE provides higher electrical efficiency and the opportunity of heat integration with the downstream synthesis process [10–12], which can significantly enhance the overall system performance. SOE co-electrolysis was initially proposed for O₂ supply and propulsion in spacecraft by the National Aeronautics and Space Administration in the 1960s [1]. It was then ignored for decades because of the low price of fossil fuels. Early this century, under the pressure of climate change and fossil fuel shortages, SOE regained global attention [1]. In particular, SOE co-electrolysis technology to produce syngas has attracted the interest of researchers. By far, a large number of papers about the materials and performance of SOE co-electrolysis have been published. The initial performance and durability of a SOE applied to

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co-electrolysis of CO_2 and H_2O were examined in Refs. [13], where the co-electrolysis performance is slightly lower than that of steam electrolysis, but is much higher than that of CO_2 electrolysis. The results of [14] shows that co-electrolysis effectively increases the syngas yield for a selected H_2O and CO_2 process stream, which appears to be a promising technique for large-scale syngas production from nuclear and renewable energy. Co-electrolysis of H_2O and CO_2 was studied in SOE [15], to demonstrate that electrochemical reduction of both CO_2 and H_2O occurs in SOE. The LSCM-CU|LSGM|LSCF SOE cells were prepared and characterized [16], and the experimental results showed better electrochemical performance for co-electrolysis of H_2O and CO_2 because of Cu impregnation [17]. Both papers focused on the production of syngas, using co-electrolysis in SOE, and studied the production of methane internally at high pressure conditions.

It has been proven that co-electrolysis of H_2O and CO_2 in SOE can produce a syngas suitable for the synthesis of chemical products, and can be considered a power-to-chemicals process (PtC). The most common products of PtC processes include, among others, methane, methanol, dimethyl ether, formic acid, and Fischer–Tropsch (FT) fuels [18–20]. The SOE co-electrolysis process was proposed in Refs. [20] to further produce liquid fuels through the FT process, revealing that the energy conversion efficiency of the SOE co-electrolysis is about 10–15% higher than a low-temperature water electrolyzer system. The authors of [21] also integrated the Fischer–Tropsch process with SOE co-electrolysis, and the designed system achieved an overall energy efficiency of 51% (LHV) when operating the SOE at 1.6 bar which is 2.5% higher than that of operating the SOE at 5 bar. Similar process has also been studied in Ref. [22], the power-to-Fischer-Tropsch fuel efficiency was estimated to be 57.2% (LHV) in the best scheme. In [23] co-electrolysis and the methanation process were integrated, and the influence of operation pressure on internal methanation was discussed. The system efficiency is higher than that of steam electrolysis (83% vs 79% LHV). In [24] the methanol and dimethyl ether synthesis process were integrated, concluding that CO-based syngas production via high-temperature co-electrolysis of H_2O and CO_2 , or alternatively high-temperature CO_2 electrolysis coupled with the WGS process, can be identified as the best technologies from energy consumption and CO_2 conversion points of view. In [25] CO_2 capture and methanol synthesis process were integrated raising the overall energy efficiency to 41% (LHV).

Methanol is one of the top chemicals produced worldwide [26,27] and can be used as a fuel in internal combustion engines and in direct methanol fuel cells, can be reformed to produce hydrogen, and blended with gasoline as an additive [28–30]. Methanol can also be used as a raw material to synthesize other fuels and chemicals (such as olefins, gasoline, dimethyl ether, methyl tert-butyl ether, acetic acid, and formic acid [31–33]). In 2017, there was a huge demand for methanol: 78 million tons (excluding methanol produced in integrated coal-to-olefins facilities). And the figure is expected to reach 100 million tons by 2025 [34].

The SOE co-electrolysis driven by renewable electricity is vital to the commercialization of the power-to-methanol process [35], which converts renewable electricity first into a syngas ($(\text{H}_2 - \text{CO}_2)/(\text{CO} + \text{CO}_2) = 2.05$) through the co-electrolysis of H_2O and CO_2 and then into methanol via syngas methanolization. From the process system point of view, the rational heat integration of the exothermic methanol synthesis process with the high temperature SOE process is crucial to the overall process performance, especially the section of the optimal stack operating conditions, such as operating pressure, average current density and voltage and outlet temperature. On the other hand, from the perspective of economic feasibility, SOE stack lifetime and price, and renewable electricity

price have important influence on the production cost of methanol and the project payback time. Therefore, the overall process performance and economic feasibility of the power-to-methanol process should be studied in detail from the perspective of process system engineering in a comprehensive way. This is the main aim of this paper, and, to the authors' knowledge, this topic has not been addressed similarly in previous papers

In this paper, a large-scale power-to-methanol system is designed with the co-electrolysis of CO_2 and H_2O in a SOE and is optimized from the techno-economical point of view. The thermodynamic performance and economic feasibility of the system are investigated. The study mainly discusses the following aspects: (1) system-level heat integration and waste heat recovery, (2) the impact of operating variables on the thermodynamic performance and economics of the system, and (3) the impact of major equipment and raw material prices on the economic competitiveness of the system.

This paper is organized as follows: Section 2 introduces the investigated PtM-CE system, including the co-electrolysis process, methanol synthesis process, heat exchanger and steam turbine network, and hot and cold utility. Section 3 describes the methodology of the proposed system and multi-objective optimization, and defines the objective functions of the system. Section 4 discusses the results in detail and analyzes the sensitivity of economic feasibility. The conclusions are presented in Section 5.

2. System description and process modeling

The proposed power-to-methanol system is designed and simulated in ASPEN PLUS with steady-state models. The plant produces annually 100 kton of methanol with a purity of 99.5 wt%, which is the common scale of commercial methanol plants. The detailed system description and process modeling are shown hereafter.

2.1. System description

The power-to-methanol system with co-electrolysis (CE) of CO_2 and H_2O in SOE (PtM-CE) is illustrated in Fig. 1. The CE process has been described in a previous authors' publication [23]. The concept assumes CE to prepare the syngas for the methanol synthesis process. The demineralized water (1) is first vaporized and mixed with gas recirculation (12) and CO_2 (6). The mixed feed (7) is further heated to 750 °C by SOE outlets, waste boiler, steam turbine extraction or electrical heating if necessary. The 750 °C feed (8) is then fed to the cathode of SOE. The produced gas mixture (9) is cooled to 40 °C before entering a flash drum, where most of the unreacted water is separated. The flow rates of CO_2 (4) and water (1) are adjusted together with the controllable operating variables of the SOE to produce an H_2 -rich syngas (19) with a modular number (defined as $(\text{H}_2 - \text{CO}_2)/(\text{CO} + \text{CO}_2)$) of 2.05 [37–39] for the methanol synthesis process. The sweep oxygen (14) is cooled or heated to 750 °C and fed into the SOE for the thermal management of the stack. Along with the production of the H_2 -rich syngas, the produced O_2 flows across the membrane and out of the anode. To enhance the production ability of the system, the sweep oxygen (14) is cooled or heated to 750 °C and fed into the SOE for the thermal management of the stack. The generated oxygen (16) is cooled to 40 °C and sold as a by-product.

The H_2 -rich syngas (19, a gas mixture of H_2 , CO, CO_2 , and a small amount of CH_4) is first compressed to 54 bar and cooled to 40 °C, then mixed with the recirculated gas (36). The reactant (22) is further compressed to 56 bar and heated up to 200 °C, and finally sent to the methanol reactor which operates in an isothermal mode at 56 bar and 230 °C. The gaseous stream (25) leaves the reactor at

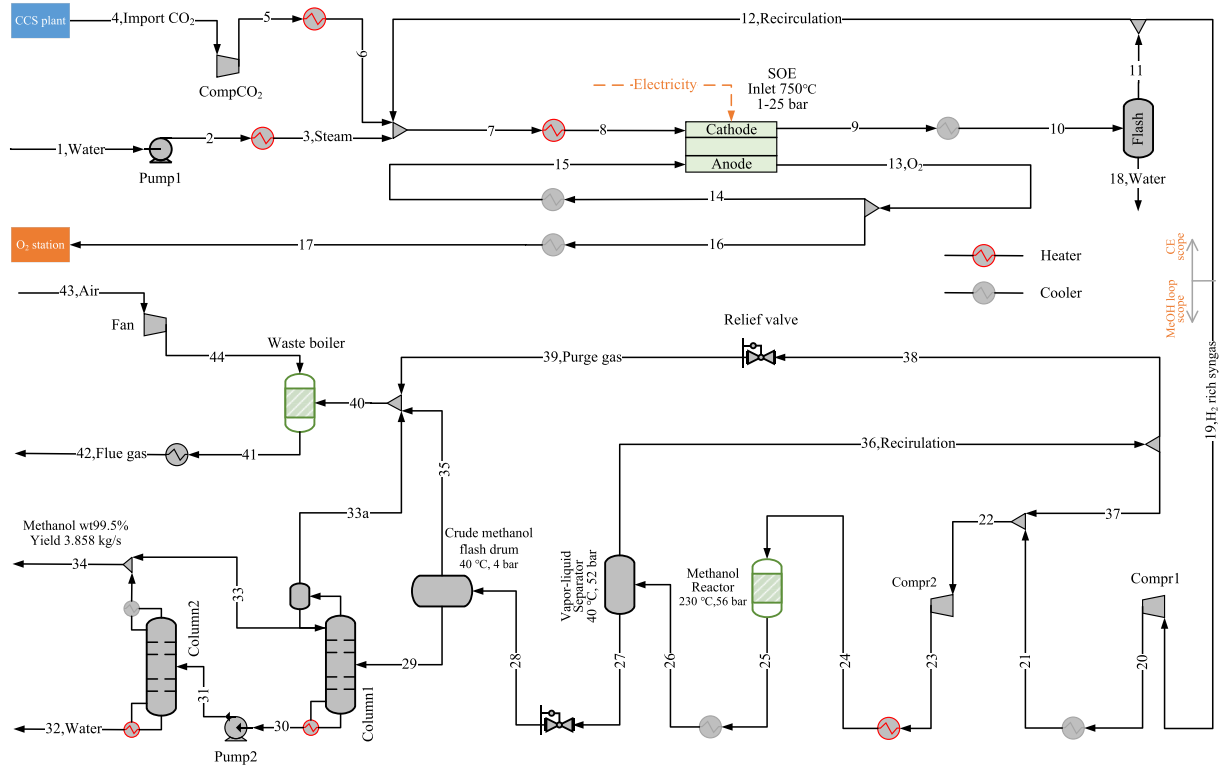


Fig. 1. Schematic of power-to-methanol with co-electrolysis of CO₂ and H₂O in SOE. The heat exchanger network is not explicitly designed but its performance is estimated with the classical chemical engineering method in Ref. [36]. The steam Rankine cycle (steam turbine network) is not explicitly presented but is optimally placed and sized for heat recovery.

230 °C. Then, it is cooled to 40 °C and fed (26) to a vapor-liquid separator at 52 bar, where the gas and liquid phases are then separated. The gas stream (36) mainly consists of H₂, CO, CO₂, CH₄ and trace amount of methanol in the vapor-liquid separator. It is compressed and recycled back to the methanol reactor. A small amount of the recycled gas (36) is purged to avoid the accumulation of inert gases and CH₄. The liquid stream (27) in the vapor-liquid separator is depressurized to 4 bar and fed (28) to the crude methanol flash drum. The released gas (35) of the crude methanol flash drum is mainly composed of CO₂, H₂, CO, methanol and CH₄, and is fed to a waste boiler for heat recovery with the purge gas (39). The liquid phase of the crude methanol flash drum is an almost gas-free mixture of methanol and water, with the concentration of methanol being around 91 wt%. The crude methanol (29) is then purified to obtain pure methanol (99.5 wt%) in two distillation columns.

2.2. Process modeling

The key components to be modelled for the system described in section 2.1 are (1) the SOE, (2) methanol synthesis reactor and distillation for methanol upgrading, and (3) heat-exchanger and steam-turbine networks.

2.2.1. Solid-oxide co-electrolysis process

The state-of-the-art electrode-supported solid-oxide electrolyzer operates at the temperature of 650–850 °C and consists of a porous Ni-YSZ (Nickel-Yttria-Stabilized Zirconia) cathode (fuel electrode), a dense electrolyte layer based on YSZ, and a porous LSM-YSZ (Lanthanum Strontium Manganite-Yttria-Stabilized Zirconia) anode (oxygen electrode) [40].

The interfaces between the electrolyte and electrodes form the so-called triple phase boundaries (TPB), where the electrochemical

reactions occur. For the CE, both H₂O and CO₂ are fed to the electrolyzer and are directly converted to H₂, CO, and O₂. The H₂O and CO₂ molecules diffuse through the cathode of the gas channel and eventually reach the cathode TPB, where the H₂ and CO reactions (Eq. (1) and (2)) take place [21,41].



The generated H₂ and CO diffuse through the cathode and returns to the gas channel, while the O²⁻ ions are selectively transported across the electrolyte and subsequently oxidized to O₂ molecules at the anode TPB (Eq. (3)) [41]. Similarly, the generated O₂ diffuses through the anode and reaches the corresponding gas channel. When the generated H₂ and CO flow through the cathode, the reverse water-gas shift and methanation reaction (internal methanation) occur with the presence of the Nickel-based catalyst as given by Eqs. (4) and (5) [21,41].

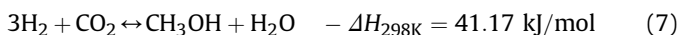
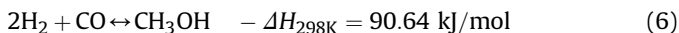


A Quasi-2D SOE model using electrochemistry, mass diffusion, heat transfer, reaction kinetics and chemical equilibrium was developed and improved in previous works [23,41]. The SOE model is calibrated using the test data of cell and stack, so it can reasonably predict the SOE performance of SE or CE under various operating conditions. The SOE inlet temperature is fixed at 750 °C, and the maximum temperature gradient between the inlet and the

outlet is 120 °C. The reactant feed is heated up to 750 °C by the SOE outlet gases, the excess heat of methanol synthesis and additional electrical heating (if the whole system still has a heat gap).

2.2.2. Methanol synthesis and purification process

Methanol synthesis is an exothermic process by means of two important reactions (Eq. (6) and (7)) in the methanol reactor (200–300 °C, 50–100 bar) with Cu/ZnO/Al₂O₃ as the catalysts [42–44].



The reactions show that during the methanol synthesis process, the H₂ consumption of CO methanolization is 66% of that of CO₂ methanolization to synthesize the same amount of methanol. And the former is more exothermic, which means that CO is driving the thermodynamic equilibrium to a higher conversion [35]. Therefore, the syngas required to synthesis methanol expects more CO and less CO₂. The commercial methanol synthesis process normally uses syngas from coal gasification and methane reforming, and the optimum module number $M = (\text{H}_2 - \text{CO}_2)/(\text{CO} + \text{CO}_2)$ of the syngas should be controlled at 2.0–2.05 [46]. Alternatively, the syngas required for methanol synthesis can also be produced from the co-electrolysis of CO₂ and H₂O in SOE [35]. A large amount of heat is released from methanol reactor and can be integrated with the SOE and methanol purification processes, thus significantly enhance the overall system performance.

A low-pressure methanol synthesis and purification model was developed in detail and key technology specifications referred to industry data in our previous work [45,46]. A methanolization reactor (stoichiometric model) at isothermal conditions at 56 bar and 230 °C was employed, and the distillation column modelled with a RadFrac model. The waste boiler was modelled by an RGibbs model under isothermal conditions at 0.2 bar and 1000 °C. The remained components (e.g., pump, compressor, and flash drum/vapor-liquid separator) were modelled in classical ways of chemical engineering.

2.2.3. Heat exchanger and steam turbine network

The performance of the heat exchanger network is estimated with heat cascade calculations as described in Refs. [47–49], the minimum temperature differences were 10 °C for liquid streams, 20 °C for low-temperature gas streams, and 30 °C for high-temperature gas streams; the excess heat is recovered in the steam turbine network to increase the overall thermodynamic performance, and dealt with by the superstructure-based framework developed at EPFL, as described in Ref. [48–50].

2.2.4. Utility

In the study, electrical heating is adopted as the hot utility to close the energy balance of the whole system; and the cooling water from rivers or lakes is used as the cold utility. The energy efficiency of electrical resistance heating is assumed to be 100%. The volume of the cooling water is assumed to be unlimited at 20 °C [23].

3. Optimization methodology and specifications

In this study, techno-economic evaluation and multi-objective optimization (MOO) are carried out on the OSMOSE platform [51–54] developed by the Group of Industrial Process and Energy Systems Engineering at EPFL, which can integrate the Aspen models developed and perform plant-wise heat integration to close

Table 1

Selected decision variables and relevant bounds.

Variables	Lower bound	Upper bound	Value	Unit
SOE ^a				
SOE operating pressure	1.1	25	–	bar
SOE utilization factor ^b	0.3	0.8	–	–
Steam feed flow rate	0.5	20	–	sccm/cm ²
Sweep feed flow rate	0.1	40	–	sccm/cm ²
Steam turbine				
Header pressure	10	98	–	bar
Super-heating	1	250	–	°C

^a The SOE inlet temperature is set at 750 °C, the maximum temperature gradient between the inlet and the outlet is 120 °C.

^b The SOE utilization factor is defined as the share of the inlet reactants (H₂O and CO₂) converted electro-chemically, which is also referred to as electro-chemical utilization factor [55].

the energy balance of the overall system. With the methodology illustrated in detail in Refs. [41,52–54], the platform has been adopted to deal with the optimization problems of various energy systems. Heat cascade utilization, formulated in mixed integer linear programming problem, is calculated to select and size hot and cold utilities. With the sized utilities, the minimum number, area and cost of the heat exchangers are estimated by the vertical heat transfer. MOO is conducted using evolutionary algorithms with the variables listed in Table 1. The optimization problem is to be investigated via two objective functions—energy efficiency and methanol production cost described hereafter.

3.1. Thermodynamic performance

The thermodynamic performance is evaluated by using energy (η) and exergy efficiency (ϵ):

$$\eta = \frac{\dot{M}_{\text{MeOH}} \cdot \text{LHV}_{\text{MeOH}} + \dot{\Delta E}^-}{\dot{\Delta E}^+} \cdot 100 \quad (8)$$

$$\epsilon = \frac{\dot{M}_{\text{MeOH}} \cdot E_{\text{MeOH}} + \dot{\Delta E}^-}{\dot{\Delta E}^+} \cdot 100 \quad (9)$$

where $\dot{M}$ is the mass flowrate and $\dot{E}$ is the exergy rate. The superscripts – and + refer to the net produced (output to power grid) and consumed (input from power grid) services, respectively [56].

The overall carbon conversion efficiency is calculated in Eq. (10):

$$c_C = \frac{\dot{m}_{\text{C,out}}}{\dot{m}_{\text{C,in}}} \cdot 100 \quad (10)$$

where $\dot{m}_{\text{C,in}}$ and $\dot{m}_{\text{C,out}}$ are the molar flowrates of carbon dioxide at the inlet and methanol product at the outlet, respectively.

3.2. Economic performance

The economic assumptions are listed in supplement material Part A. The performance indicators of the economic evaluation are investment cost, methanol production cost, and payback time. The capital expenditure (CAPEX) is estimated based on [36] with the uncertainty up to $\pm 30\%$. A capacity factor is considered to estimate the CAPEX of the components (e.g., methanol reactor, steam turbine network, and SOE) by scaling up from base components, the base cost from Ref. [57–59], with the parameters given in supplement material Part A. For the remaining components, such as pump, compressor, heat exchanger, flash drum, distillation column and waste boiler, additional pressure and material factors are

involved on the basis of [36] and listed in supplement material Part A.

With the calculated investment cost, the depreciation cost can be computed by Eq. (11) [37]:

$$C_{\text{dep}} = C_{\text{inv}} \times \frac{i \times (i + 1)^n}{(i + 1)^n - 1} \quad (11)$$

where C_{inv} is investment cost, i is annual interest rate, and n is plant lifetime (year). The methanol production cost (\$/ton) is defined by Eq. (12) [37]:

$$C_{\text{MeOH}} = \frac{C_{\text{opt}} + C_{\text{dep}} - C_{\text{rev}}^{\text{byp}}}{P_{\text{MeOH}}} \quad (12)$$

where P_{MeOH} is the capacity of methanol production (ton/year) and $C_{\text{rev}}^{\text{byp}}$ is byproduct revenues from the sale of the exported electricity, oxygen, and carbon dioxide.

The payback time τ is defined to be the total investment cost divided by the annual profit of the methanol plant in Eq. (13) [37]:

$$\tau = \frac{C_{\text{inv}}}{C_{\text{MeOH}}^{\text{rev}} + C_{\text{rev}}^{\text{byp}} - C_{\text{opt}}} \quad (13)$$

where C_{opt} is operation cost that includes labor, biomass, catalysts and imported electricity (\$/year), and $C_{\text{MeOH}}^{\text{rev}}$ is methanol revenues (\$/year).

4. Results and discussion

The PtM-CE case is techno-economically optimized with a fixed scale of 1×10^5 ton of methanol per year by means of Pareto-optimal solutions (namely Pareto front) shown in Fig. 2. There is a slight trade-off between energy efficiency (LHV) and methanol production cost. When energy efficiency increases from 71.58% to 72.0%, methanol production cost rises by merely 6 \$/ton. Seen from the optimization results, there is almost an optimal solution for the designed PtM-CE case. In this paper, the maximum efficiency point (MEP) is seen as the research object and the main results are summarized in Table 2, where the optimized PtM-CE system: (1) achieves a higher system energy efficiency (72.0%) but results in a longer payback time longer than 11 years; (2) annual utilization of carbon dioxide is 146.7 kton, and the carbon conversion efficiency is as high as 93.6%; (3) has an energy storage capacity of 100 MW;

Table 2
Summary of results of different design solutions.

	Items	Unit	Value
Consumption	CO ₂	kton/year	146.7
	H ₂ O	kton/year	163.2
	Electricity (SOE)	MW	103.8
	Electricity (others)	MW	6.2
Production	MeOH	kton/year	100
	O ₂	kton/year	162.8
	Electricity	MW	3.37
Efficiency	η	%	72.0
	ϵ	%	76.8
	C_C	%	93.6
	C_{inv}	M\$	150.4
Economic parameters	C_{opt}	M\$/year	65.9
	C_{MeOH}	\$/ton	537
	τ	years	11.3

and (4) can co-produce methanol, oxygen (162.8 kton/year), and electricity (3.37 MW).

4.1. Analysis of thermodynamic performance

Table 3 shows the decision variables at the MEP of the Pareto front and the main operational values at the MEP. The optimal SOE operating pressure of PtM-CE is close to the lower bound of the decision variable in Table 1, because the nickel-based electrode promotes internal methanation, which quite favorable along with the increase of the operating pressure. The effects of co-electrolysis operating pressure on internal methanation has been studied in detail in author's previous work [23], e.g., when SOE operates under low pressure (1 bar with 50% utilization factor), the amount of methane produced is close to zero. By increasing the pressure up to 30 bar the methane fraction reaches 7% vol% dry gas, which in turn benefits the methane producing system. However, for methanol synthesis process integrated with co-electrolysis, SOE is more suitable for low-pressure operation to avoid internal methanation of cathode. Based on the decision variables of optimal configuration in Table 3, the impact of SOE operating pressure on cathode outlet component, energy efficiency and electrolysis power is shown in Fig. 3 with additional information given in Part B of the supplement material. Fig. 3a shows the variation trend of the concentration of SOE cathode outlet components (dry basis) along with the increase of SOE pressure. It indicates that with the increase of operating pressure, the concentration of methane at the cathode outlet

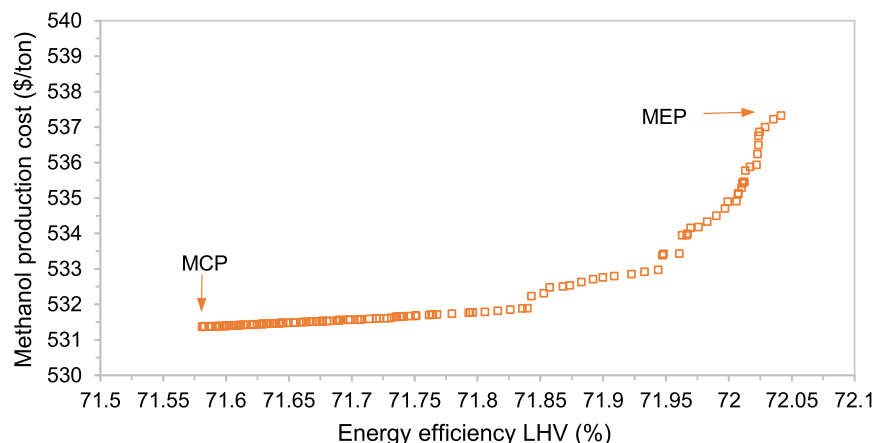


Fig. 2. Trade-off between methanol production cost and system energy efficiency.

Table 3
Decision variables and main operational values of the optimized configuration.

	Items	Unit	Value
SOE	Operating pressure	bar	1.31
	Inlet temperature	°C	750
	Steam feed flowrate	scm/cm ²	5.9
	CO ₂ feed flowrate	scm/cm ²	2.2
	Sweep feed flowrate	scm/cm ²	0.1
	Average current density	A/cm ²	1.0
	Average cell voltage	V	1.37
	Utilization factor	—	0.8
	Outlet temperature	°C	861
	Cathode outlet composition (H ₂ /CO ₂ /CO/CH ₄)	% (dry basis)	68.7/4.8/26.4/0.1
	Syngas module number	—	2.05
	Stack power	kW/stack	7.06
	Stack number	—	14,714
	Operating pressure	bar	56
Methanol synthesis	Operating temperature	°C	230
	Header pressure	bar	98
	Header temperature	°C	554
	Condensation	bar/°C	0.05/28

increases obviously, while the concentration of H₂ and CO decreases (When the pressure increases to 25 bar, the concentration of SOE cathode outlet composition (dry basis) is H₂/CO₂/CO/CH₄ = 65.2%/5.2%/23.9%/5.6%). This further indicates that the production of the same amount of methanol requires a larger SOE size to meet the H₂ and CO consumption of methanol synthesis. However, the amount of electricity required for electrolysis increased significantly, resulting in a gradual decrease in overall energy efficiency (when the pressure increases to 25 bar, the energy efficiency drops to 63.4%), as shown in Fig. 3b.

The Carnot composite curve of system-level heat integration and system performance of PtM-CE is shown in Fig. 4, and the corresponding operating parameters are listed in Table 3. In addition, more composite curves are reported in Part C of the

supplement material. As it can be seen from the Carnot grand composite curve (Fig. 4a), the optimal configuration is obtained through techno-economic optimization, the SOE exothermic model (with the outlet temperature being 861 °C) is preferred, the system can realize the self-sufficient heat management without additional hot utility, and the excess heat is recovered for the co-production of electricity by the steam turbine network to enhance the system efficiency. The segment from left to right means excess heat that needs to be extracted from the system, whereas the system needs to absorb heat. Above 220 °C, there is a large energy pocket, the heat released from waste boiler and SOE outlets is utilized for HP steam generation and superheated steam in the steam turbine network. Below 220 °C, there is another energy pocket. A significant heat demand for the steam generation of SOE and two-stage distillation reboiler is satisfied by the methanol reactor and steam extraction after expansion in steam turbine.

Further insights about the integration are reflected by the integrated Carnot composite curve of SOE in Fig. 4b: (1) part of the high-temperature heat stream at the SOE outlets heat the inlet stream to 750 °C, (2) the remaining high-temperature heat at the SOE outlets and heat released from the waste boiler are recovered by the steam turbine network, (3) a large amount of exothermic heat from the methanol reactor is used to steam generation of SOE, (4) the extracted steam of the steam turbine network and the low temperature portion of excess heat (SOE outlets, methanol reactor outlet and flue gas) are used to heat the reboiler of the two-stage distillation column.

4.2. Evaluation of economic performance

4.2.1. Cost distribution

Taking capital and operational expenditure into consideration, the key indicators of economic evaluation are investment cost, methanol production cost, and payback time. Assuming a 25 years plant lifetime [37], 10% interest rate [37], 7200 annual operating hours (AoHs) [60], 2000 \$ of single stack price [59], and a stack

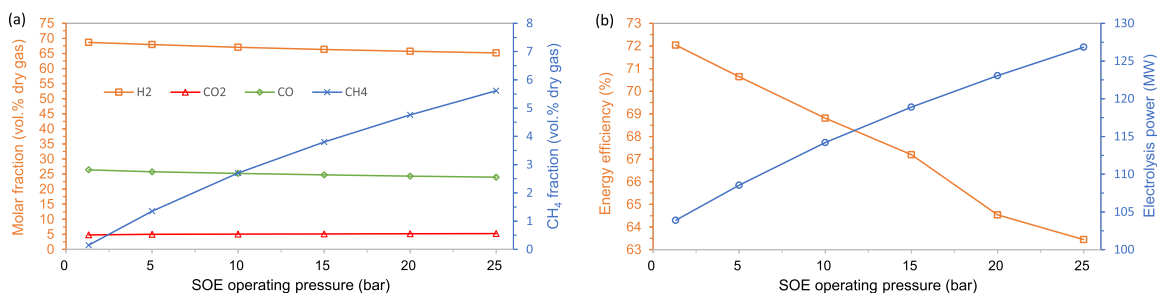


Fig. 3. Impact of SOE operating pressure: (a) Cathode outlet composition, (b) Energy efficiency and electrolysis power.

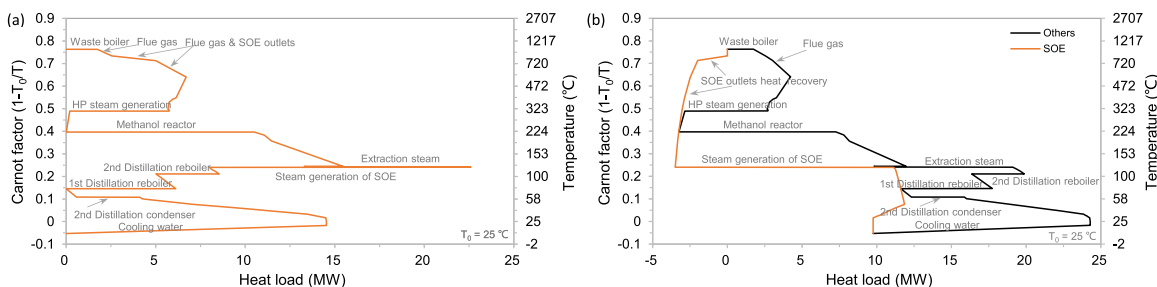


Fig. 4. The Carnot composite curve of system-level heat integration and system performance: (a) Carnot grand composite curve of PtM-CE, (b) Integrated Carnot composite curve of SOE.

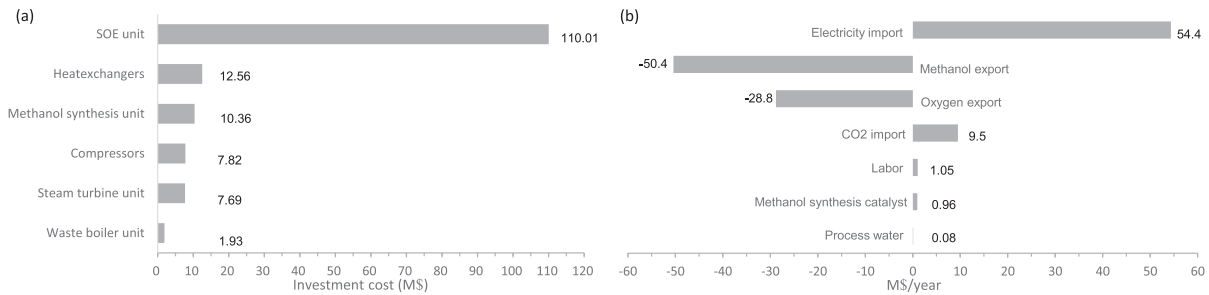


Fig. 5. Cost distribution of the scenarios defined in Table 3 with a plant capacity of 100 kton of methanol per year: (a) Investment distribution, (b) Operation and revenue distribution.

lifetime of 48,000 h [61], the stacks will be replaced 3 times for the whole plant lifetime, and the replacement cost is included in capital investment. More detailed economic assumptions and capital expenditure are given in supplement material Part A. The major investment units of the PtM-CE plant include SOE, heat exchangers, compressors, methanol synthesis, steam turbine, and waste boiler. As shown in Fig. 5a, the SOE unit (110.01 M\$) accounts for 73% of the total investment (150.4 M\$), much more expensive than the other units. The investment into heat exchangers is about 12.56 M\$ (8.4% of the total); and that in the methanol synthesis unit is about 10.37 M\$ (6.9% of the total). The unit consists of methanol synthesis reactor, flash drum, separator and distillation column. The investment in the steam turbine unit is about 7.7 M\$ (about 5.1% of the total); and that into the waste boiler unit is about 1.93 M\$ (1.3% of the total). It is clear that SOE's investment dominates the total investment of the project. Fig. 5b shows the distribution of operation cost and revenue of the PtM-CE plant, where positive and negative values represent operation cost and revenue, respectively. The operation cost is about 65.99 M\$/year, covering the imported electricity cost (54.4 M\$/year), the imported CO₂ cost (9.5 M\$/year), the labor cost (1.05 M\$/year), the catalyst cost (0.96 M\$/year), and the process water cost (0.08 M\$/year). The revenue is about 79.2 M\$/year, including the export of methanol (50.4 M\$/year) and oxygen as a by-product (28.8 M\$/year). The cost of imported electricity accounts for 82.4% of the total operation cost and is 1.1 times of that of the methanol revenue. The cost of the imported CO₂ comes next, accounting for 14% of the total operation cost. According to the calculation method in Section 3.2, the methanol production cost and project payback time is 537 \$/ton of methanol and 11.3 years respectively. Obviously, it is financially not competitive. The main influence factors of economic feasibility are the prices of the SOE stack, imported electricity, and the imported CO₂. In addition, as mentioned in the introduction, one of the main purposes of the designed system is to solve the storage problem of renewable electricity. In actual operation, due to the instability of renewable electricity, its annual available hours will be lower than the assumed annual operating hours (7200), which will significantly affect the economic feasibility of the project. Detailed sensitivity analysis has been carried out, as shown in Section 4.2.2.

4.2.2. Sensitivity analysis of economic performance

To further study the impact of the prices of SOE stack, imported electricity, imported CO₂, and AoHs on economic feasibility, a sensitivity analysis is carried out. At present, SOEs do not have a market price because they are still at laboratory stage. In this study, the purchase price of SOE stack is assumed to be 2000 \$/stack, with its lifetime being 48,000 h. The price is expected to fall to a great extent along with the maturity of technology and scale production. The stack lifetime (48,000 h) is below the lower bound (60,000 to

90,000 h [62]) put forward by industry experts. Therefore, SOE stacks need to be replaced for three times when the project lifetime is 25 years and can be operated for 7,200 h per year. In the future, if the lifetime of SOE stacks is doubled to 96,000 h, SOE stacks will only need to be replaced once, which may significantly reduce the investment cost of SOE. Therefore, the sensitivity analysis adopts a price range of 500–2,000 \$/stack with the stack lifetime being 48,000 and 96,000 h. The wholesale price of electricity differs from country to country. For example, the price in the 4th quarter of 2017 in European countries are: 62 €/MWh in Italy, 61 €/MWh in Portugal and Greece, 60 €/MWh in Switzerland, 57 €/MWh in France, 31 €/MWh in Denmark and Sweden, and 30 €/MWh in Norway [62]. The sensitivity analysis was performed in the range of 20–60 €/MWh. CO₂ is one of the major raw materials for the PtM-CE process, and its annual imported cost accounts for about 14% of the annual operation cost, second only to the cost of imported electricity. As CCS technology advances and the decrease of CO₂ transportation cost of, the price of imported CO₂ will be lower. Hence, the sensitivity analysis assumes the purchase price of CO₂ to be 20–50 €/ton. In this paper, it is assumed that the PtM-CE plant has an AoHs of 7200 according to the design regulations of the commercial methanol plant. For the SOE driven by renewable electricity, the plant needs to be shut down when renewable electricity is not available. It means that AoHs of PtM-CE plant is usually less than 7200 h due to the instability of renewable electricity (wind and solar PV). The sensitivity analysis of AoHs was performed in the range of 1800–7200 h/year.

The results of the sensitivity analysis are summed up in Fig. 6. As shown in Fig. 6a, with the SOE stack lifetime being 48,000 h, when the stack price decreases from 2,000 to 500 \$/stack, the methanol production cost can be reduced from 537 to 447 \$/ton with the payback time reduced from 11.4 to 5.1 years accordingly. If the stack can last for 96,000 h, the methanol production cost will be reduced from 500 to 437 \$/ton with the payback time reduced from 8.8 to 4.5 years accordingly. Fig. 6b is the sensitivity analysis of the imported electricity price: when the price drops from 60 to 20 €/MWh, the methanol production cost can be reduced from 537 to 157 \$/ton with the payback time reaching 3 years. Fig. 6c is the sensitivity analysis of the imported CO₂ price: when the price drops from 60 to 20 €/MWh, the methanol production cost can be reduced from 537 to 486 \$/ton with the payback time being 8.1 years. Fig. 6d is the sensitivity analysis of the AoHs: when the AoHs drops from 7200 to 3600, the methanol production cost increase significantly from 537 to 649 \$/ton with the payback time being 21 years, which is very close to the project lifetime (25 years).

From the above analysis, it can be seen that methanol production cost and project payback time are greatly influenced by the SOE stack price, the imported electricity price, and AoHs. The development of SOE in financial term and low electricity market

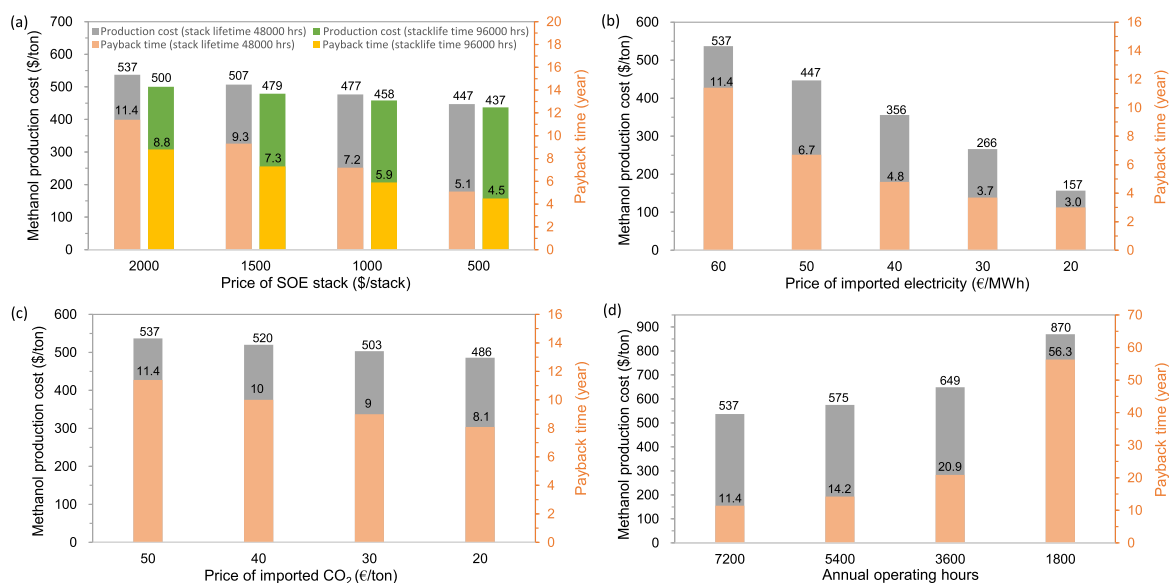


Fig. 6. Sensitivity analysis of main variables: (a) SOE stack price 500–2000 \$/stack with stack lifetime being 48,000 and 96,000 h, (b) Electricity purchase price 20–60 €/MWh, (c) CO₂ purchase price 20–50 €/ton, (d) annual operating hours 7200–1800.

price greatly reduce methanol production cost and expects a payback time of 3–5 years. In particular, when considering the imported electricity price from the perspective of electricity storage, the large variation due to electricity market plays a decisive role in implementing such integrated plant concepts under the current SOE price. CO₂ price also influences the economy of the system, but it is less important than electricity price and SOE purchase cost. The annual available hours of renewable electricity have a significant impact on the economic feasibility of PtM-CE plant. A stable supply of renewable electricity is a prerequisite for PtM-CE's investment.

5. Conclusion

In this study, the techno-economic optimization of power-to-methanol with co-electrolysis of CO₂ and H₂O in a solid-oxide electrolyzer is carried out. The system is first designed in detail and the model is developed using ASPEN Plus. Then, a multi-objective optimization and system-level heat integration were done to analyze the performance of the optimized conceptual designs in terms of energy efficiency and methanol production cost. A sensitivity analysis of three influential factors is performed in the end to figure out the main factors affecting the payback time. Major conclusions are as follows:

- There is a slight trade-off between energy efficiency (LHV) and methanol production cost for the power-to-methanol system with co-electrolysis of CO₂ and H₂O in solid-oxide electrolyzer. The proposed system achieves a higher system energy efficiency (72.0%) and carbon conversion efficiency (93.6%) but results in high methanol production costs (537 \$/ton) and long payback time of over 11 years.
- The proposed system is equipped with both high capacity for renewable electricity storage and carbon dioxide utilization as well as the capacity of co-producing oxygen (162.8 kton/year) and electricity (3.37 MW). The annual utilization of carbon dioxide is 146.7 kton, and the capacity for renewable energy storage can reach 100 MW.

- Since internal methanation can be effectively promoted by enhancing the operating pressure, for the co-electrolysis integrated methanol synthesis process, SOE is more suitable to low-pressure operation to avoid internal methanation of cathode.
- The optimized system has perfect overall heat integration property. SOE prefers the exothermic model (with the outlet temperature being 861 °C) and heats the feed steam and CO₂ to 750 °C through the outlets. A large amount of exothermic heat from methanol reactor can meet the steam generation demand of SOE without the intervention of external hot-utility. The excess heat is recovered by the steam turbine network for electricity co-production.
- The proposed case is financially not competitive when the stack price is 2000 \$/stack, the lifetime is 48,000 h, and the electricity price is 62 €/MWh. The sensitivity analysis shows that reducing the price of SOE stack, extending its lifetime and reducing the electricity price have significant impact on project implementation and can even achieve a shorter payback time of 3–5 years. A stable supply of renewable electricity is a prerequisite for project investment. When the annual operating hours drops from 7200 to 3600, the payback time of the project increased to 21 years, which is very close to the project lifetime (25 years).

Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Hanfei Zhang: Conceptualization, Methodology, Software, Validation, Investigation, Data curation, Writing - original draft, Visualization. **Umberto Desideri:** Conceptualization, Writing - original draft, Supervision, Project administration.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.energy.2020.117498>.

Nomenclature

Abbreviations

AE	alkaline electrolyzer
AMPL	a mathematical programming language
CCU	carbon capture and utilization
CCUS	carbon capture, utilization and storage
CE	CO-electrolysis
EES	electrical energy storage
HEN	heat exchanger network
HHV	higher heating value
IEA	international energy agency
IRENA	international renewable energy agency
LCI	lifecycle indicator
LHV	lower heating value
MCP	minimum cost point
MEP	maximum efficiency point
MILP	mixed integer linear programming
MSS	methanol synthesis system
MOO	multi-objective optimizer
PEC	purchase equipment cost
PEME	polymer electrolyte membrane electrolyzer
PtH	power-to-hydrogen
SE	steam electrolysis
SOCE	solid-oxide CO-electrolysis
SOE	solid-oxide electrolysis
USD	united states dollar

Greek Symbols

η	energy efficiency
ϵ	exergy efficiency
τ	payback time

Mathematical Symbols

CAPEX	capital expenditure
C_C	overall carbon conversion efficiency
C_{dep}	depreciation cost
C_{inv}	investment cost
C_{opt}	operation cost
C_{rev}^{meth}	methanol revenue
C_{rev}^{byp}	byproduct revenue
C_{meth}	methanol production cost
E_{MeOH}	exergy rate of methanol
$\Delta \dot{E}^+$	net electric power input
$\Delta \dot{E}^-$	net electric power output
ΔH_{298K}	standard enthalpy of formation, kJ mol ⁻¹
i	annual interest rate
M	syngas modular
$\dot{M}_{MeOH}$	the mass flow of obtained methanol

n	the number of electrons exchanged in the reaction
OPEX	operational expenditure
P_{meth}	the capacity of methanol production
T	temperature, K

Superscripts

n	project lifetime
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